

2 marks

a) Define

i) Data Abstraction

ii) Instance.

Ans. i) Data Abstraction:

Hiding complexity of data structures from end user through different levels is known as data abstraction.

Many end users are not computer trained so it is needed to hide complex data structures from them.

It has 3 levels :

- a. Physical level
- b. logical level
- c. view level

ii) Instance:

The data stored in database at a particular moment of time is called instance of database.

Example:

Let's say a table teacher in our database whose name is School, suppose the table has 50 records so the instance of the database has 50 records for now and tomorrow we are going to add another fifty records so tomorrow the instance have total 100 records.

This is called an instance.

b) State any two advantages of DBMS.

Ans. Reduction in Data redundancy

Data consistency and integrity

Data security

Privacy

Easy access of data

Easy recovery

Flexibility

c) Define Normalization. Enlist its types.

Ans. Normalization is a process of organizing the data in database to avoid data redundancy, insertion anomaly, update anomaly & deletion anomaly.

Types of normalization are:

- First normal form(1NF)
- Second normal form(2NF)
- Third normal form(3NF)
- Boyce & Codd normal form (BCNF)
- Fourth normal form(4NF)

d) Write syntax for creating and Renaming a table.

Ans. Syntax of the CREATE TABLE statement is as follows:

CREATE TABLE table_name

(

column1 datatype (size),

column2 datatype(size),

column3 datatype(size));

Syntax of RENAME TABLE statement is as follows:

RENAME old_table _name To new_table_name ;

e) Enlist arithmetic and logical SQL operators.

Ans. SQL Arithmetic Operators:

Addition Operator (+)

Subtraction Operator (-)

Multiplication Operator (*)

Division Operator (/)

Modulus Operator (%)

SQL Logical Operators:

ALL operator

AND operator

OR operator

BETWEEN operator

IN operator

NOT operator

ANY operator

LIKE operator

f) Write syntax for creating and dropping views.

Ans. CREATE VIEW Syntax:

CREATE VIEW view_name AS

SELECT column1, column2...

FROM table_name

WHERE condition ;

DROP VIEW Syntax:

DROP VIEW view_name;

g) State two advantages of PL/SQL

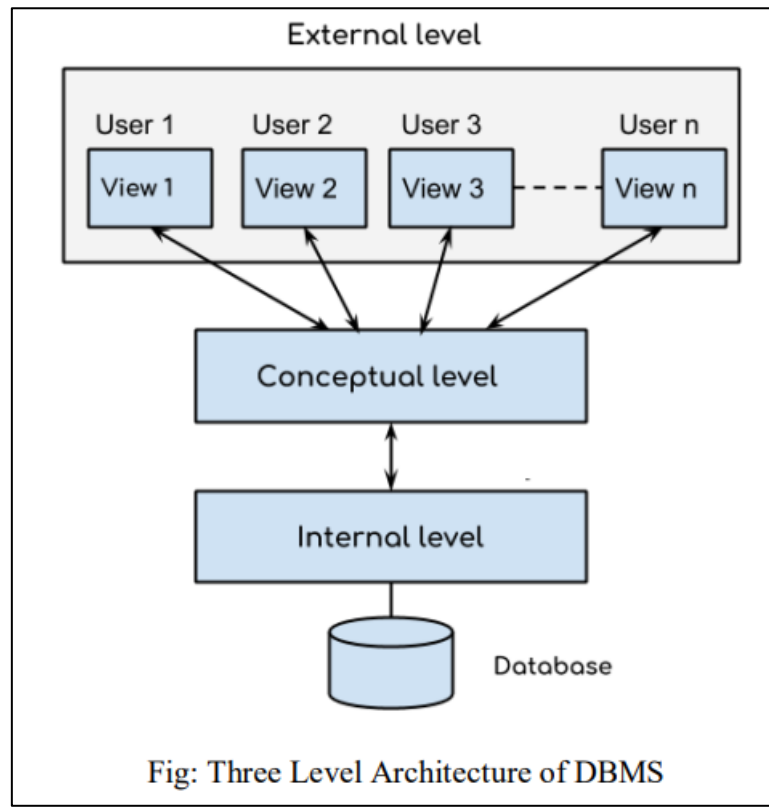
Ans. Advantages of PL/SQL are:

1. Work can be divided into smaller modules so that it can be manageable and also enhances the readability of the code.
2. It promotes reusability.
3. It is secure, as the code is in the database and hides the internal database details from the user.
4. It improves performance against running SQL queries multiple times.

4 marks

a) Explain three level architecture of Database system.

Ans.



This architecture has three levels:

1. External level
2. Conceptual level
3. Internal level

1. External level

It is also called view level because several users can view their desired data from this level which is internally fetched from database with the help of conceptual and internal level mapping.

The user doesn't need to know the database schema details such as data structure; table definition etc. user is only concerned about data which is what returned back to the view level after it has been fetched from database which is present at the internal level. External level is the top level of the three level DBMS architecture.

2. Conceptual level

It is also called logical level. The whole design of the database such as relationship among data, schema of data etc. are described in this level.

Database constraints and security are also implemented in this level of architecture.

This level is maintained by DBA (database administrator).

3. Internal level

This level is also known as physical level. This level describes how the data is stored in the storage devices.

This level is also responsible for allocating space to the data. This is the lowest level of the architecture.

b) Write SQL queries for following:

i) Create table student with following attributes using suitable data types. Roll no., as primary key, name, marks as not null and city.

ii) Add column Date of Birth in above student table.

iii) Increase the size of attribute name by 10 in above student table.

iv) Change name of Student table to stud.

Ans. i) CREATE TABLE Student(Rollno int PRIMARY KEY,name varchar(30) NOT NULL,marks int NOT NULL,city varchar(20));

ii) ALTER TABLE student ADD DateofBirth varchar(20);

iii) ALTER TABLE student Modify name varchar(40);

iv) RENAME Student to Stud;

c) Write and Explain the syntax for creating and dropping indexes with an example.

Ans. CREATE INDEX

The CREATE INDEX command is used to create indexes in tables. It allows duplicate values. Indexes are used to retrieve data from the database very fast. The users cannot see the indexes; they are just used to speed up searches/queries.

Syntax:

CREATE INDEX index_name

ON table_name (column1, column2, ...);

Example:

The following SQL creates an index named id_firstname on the FirstName column in the Student table:

CREATE INDEX id_firstname

ON Student (FirstName);

DROP INDEX

The DROP INDEX statement is used to delete an index in a table.

Syntax:

DROP INDEX index_name ON table_name;

Example:

DROP INDEX id_firstname ON Student;

d) Write a PL/SQL code to print reverse of a number.

Ans. declare

n number;

i number;

rev number:=0;

r number;

begin

n:=&n;

while n>0

loop

r:=mod(n,10);

rev:=(rev*10)+r;

n:=trunc(n/10);

end loop;

dbms_output.put_line('reverse is ' || rev);

end;

a) Write down any four Dr. E.F Codd's rules.

Ans. Rule 1: The information Rule: all data viewed to users.

Rule 2: Guaranteed Access Rule: all user get access to database

Rule 3: Systematic treatment of null values: null value should be empty

Rule 4: Dynamic online Catalog: record all transactions in database

Rule 5: Data Sub language rule: use only one language

Rule 6: View updating rule: table and view updated simultaneously

Rule 7: High level insert, delete and update: multiple insert delete update

Rule 8: Physical data independence: hardware change

Rule 9: Logical data independence: structure change

Rule 10: Integrity independence: store correct data

Rule 11: Distribution independence: distributed database

Rule 12: No subversion rule: no version of language used.

b) State the use of group by and order by clauses.

Ans. Group by Clause:

Group by clause is used to collect the data as multiple records and group them to produce the result.

Syntax:

SELECT column_name, function(column_name)

FROM table_name

WHERE condition

GROUP BY column_name;

Ex: select avg(sal) from emp_details group by deptno;

Order by Clause:

To view the data in sorted order, the order by clause is used.

By default, the data is sorted in ascending order.

Syntax:

SELECT expressions

FROM tables

[WHERE conditions]

ORDER BY expression [ASC | DESC];

Ex: select deptno from emp_details order by deptno;

Ex: select deptno from emp_details order by deptno desc; (for descending order)

c) Explain the steps of cursor implementation with syntax and example.

Ans. Cursor Implementation:

Steps to create Cursor:

a. Declaring cursor:

Cursor is declared in the declaration section.

Syntax: cursor <cursor_name>is<select query>;

Ex: cursor a is select ename from emp_details where empno=3;

b. Opening cursor:

After declaring the cursor, the cursor needs to open.

Syntax:

open<cursor_name>;

Ex: open a;

c. Fetching a record from cursor:

Once the cursor is declared and opened, we need to get records or rows from the cursor. These records are accessed using the FETCH statement.

Syntax: fetch <cursor_name> into <variable_list>;

Ex: fetch a into name;

d. Closing cursor:

Once the cursor is opened and processing is over, we need to close it.

Syntax: close <cursor_name>;

Ex:

close a;

Example:

declare

name emp_details.ename%type;

cursor a is select ename from emp_details where empno=3; //cursor declaration

begin

open a; //opening the cursor

loop

fetch a into name; //fetching the rows from cursor

update emp_details set comm=3000 where empno=3;

exit when a%notfound;

dbms_output.put_line('record updated');

end loop;

close a; //closing the cursor

end;

/

d) Explain acid properties of a transaction.

Ans. ACID Properties of Transaction:

1. Atomicity

2. Consistency

3. Isolation

4. Durability

1. Atomicity:

Atomicity means all the operations included in the single transaction gets executed at a time or none.

2. Consistency:

Consistency means update or edits the same data stored at different locations.

3. Isolation:

Isolation means all the transactions gets executed independent of each other.

4. Durability:

Durability means data can be saved in database permanently until user change it.

a) Describe any four responsibilities of Database Administrator.

Ans. Responsibilities of Database Administrator (DBA):

1. Schema Definition:

Database or schema can be designed or defined by DBA.

2. Creating storage structure:

DBA allocate or decide the space to store the database.

3. Create grant access methods:

Different access methods to access the database can be granted by DBA to the users.

4. Schema modification:

The database or schema which is already defined can be modified by DBA as per the requirements.

5. Granting authorization:

To access the different databases, DBA can grant the authorization to authorized users only.

6. Performance tuning:

The problems/errors arise in database accessing; can be resolved by DBA to increase the performance.

7. Regular maintenance:

DBA can monitor the transactions in database and maintain the database error free by doing the regular maintenance.

b) Explain Primary and Unique key constraint with syntax.

Ans. Primary key constraint:

Primary key constraint applied to any column can't accept the duplicate and null values.

This constraint can be applied at the time of table creation.

Syntax for Primary Key:

```
CREATE TABLE <Table_Name>
(Column1 datatype,
Column2 datatype,CONSTRAINT <Name> PRIMARY KEY (Column name)
);
```

Example:

```
CREATE TABLE CUSTOMERS
( ID INT NOT NULL,
AGE INT NOT NULL,
ADDRESS CHAR (25),
PRIMARY KEY (ID) );
OR
CREATE TABLE CUSTOMERS
(ID INT NOT NULL,
NAME VARCHAR (20) NOT NULL,
AGE INT NOT NULL,
ADDRESS CHAR (25) ,
SALARY DECIMAL (18, 2),
PRIMARY KEY (ID, NAME)
);
```

Unique key constraint:

Unique key constraint applied to any column can't accept the duplicate values (only accepts unique values).

This constraint can be applied at the time of table creation.

Syntax for Unique key:

```
CREATE TABLE <Table_Name>
(Column1 datatype,
Column2 datatype,CONSTRAINT <Name> UNIQUE KEY (Column name)
);
```

Example:

```
CREATE TABLE students
(S_Id int NOT NULL,
LastName varchar (255) NOT NULL,
FirstName varchar (255),
City varchar (255),
UNIQUE (S_Id)
)
OR
CREATE TABLE students
(S_Id int NOT NULL,
LastName varchar (255) NOT NULL,
FirstName varchar (255),
City varchar (255),
CONSTRAINT uc_studentId UNIQUE (S_Id, LastName)
);
```

c) Write and Explain the syntax for creating database trigger.

Ans. Database trigger:

Triggers can be referred as stored procedures that are fired or executed when an INSERT, UPDATE or DELETE statement is given against the associated table.

Syntax:

```
create trigger [trigger_name]
[before | after]
{insert | update | delete}
on [table_name]
[for each row]
[trigger_body]
```

Explanation of syntax:

1. create trigger [trigger_name]: Creates or replaces an existing trigger with the trigger_name.
2. [before | after]: This specifies when the trigger will be executed.
3. {insert | update | delete}: This specifies the DML operation.
4. on [table_name]: This specifies the name of the table associated with the trigger.
5. [for each row]: This specifies a row-level trigger, i.e., the trigger will be executed for each row being affected.
6. [trigger_body]: This provides the operation to be performed as trigger is fired

Example:

Given Student Report Database, in which student marks assessment is recorded. In such schema, create a trigger so that the total and percentage of specified marks is automatically inserted whenever a record is insert.

Here, as trigger will invoke before record is inserted so, BEFORE Tag can be used.

```
create trigger stud_marks
before INSERT
on Student
for each row
set Student.total = Student.subj1 + Student.subj2 + Student.subj3, Student.per
= Student.total * 60 / 100;
```


d) Write a trigger which invokes on deletion of record on emp table.

Ans. Delete Trigger: create or replace trigger trg1 before delete on emp_details
declare
begin
raise_application_error(-20000,'cannot delete the record');
end;

e) Explain Database Recovery techniques in detail.

Ans. Database Recovery Techniques:

Database recovery techniques are used to restore the original data in system from backup.

Backward and forward recovery is two types of database recovery.

Recovery Techniques:

1. Log based recovery.

2. Shadow paging recovery

3. Checkpoints

1. Log based recovery:

It records sequence of log records, which includes all activities done by database users.

It records the activities when user changes the database.

In case of database failure, by referring the log records users can easily recover the data.

2. Shadow paging recovery:

This technique is the alternative for log based recovery.

In this technique, database is divided into pages that can be stored on the disk.

The page table is used to maintain the record of location of pages.

In case of database failure, page table is used to recover the parts of database.

3. Checkpoints:

Checkpoint records all committed transactions into logs.

When system fails, it check log to determine recovery action.

6 marks

a) Draw the overall architecture of DBMS. Explain storage manager and query processor components.

Ans. Storage manager components:

1. Buffer Manager

The Buffer Manager allocates the space to the buffer to store data in it.

2. File Manager

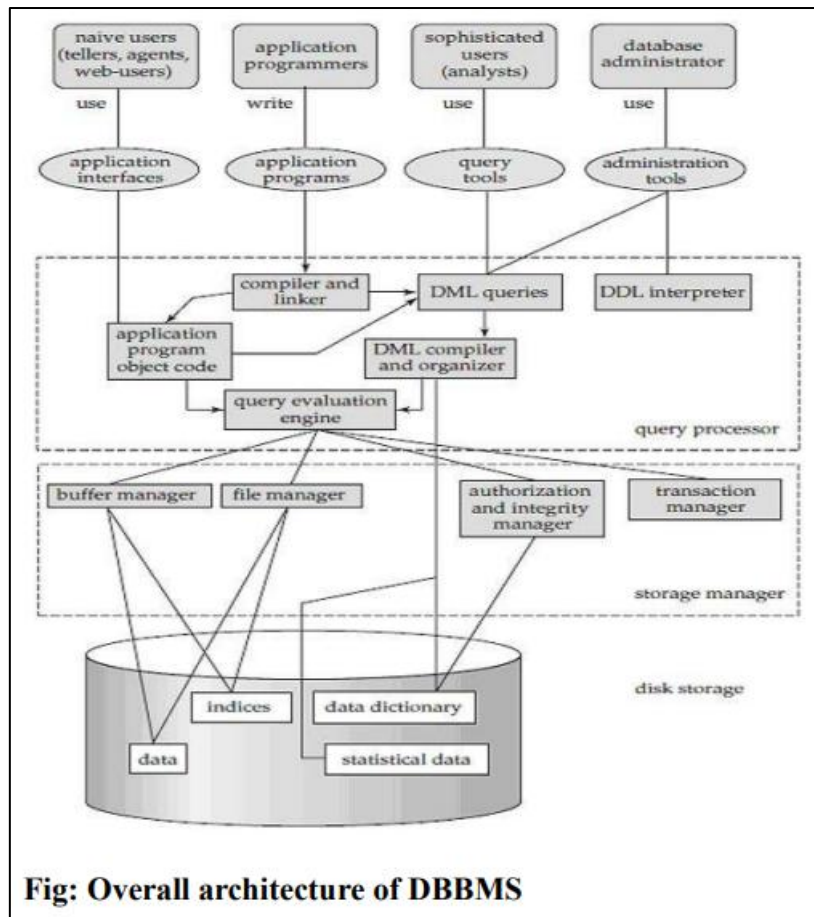
The task of the file manager is to manage the space allocation in disk for storing the information and also the data structures used for representing that information.

3. Authorization and Integrity Manager

Allows only authorized users to access data and should be hidden from the public users. The Authorization and Integrity Manager verifies the authority of the user trying to access the data and it also checks the integrity constraints when the database is modified.

4. Transaction Manager

A transaction in DBMS is nothing but a very small unit of the program. The Transaction Manager manages all the transaction (program) execution.



Query Processor Components:

Query processor handles the query processing. It processes the query given by the user.

1. DDL Interpreter:

DDL expands to Data Definition Language. DDL Interpreter interprets the DDL statements such as schema definition statements like create, delete, etc.

2. DML Compiler:

DML expands to Data Manipulation Language. DML Compiler compiles (or translates) the DML statements such as select, update and delete statements into low level instructions which is nothing but the machine-readable object code to make it executable. The DML compiler is also responsible for query optimization.

3. Query Evaluation Engine:

The Query Evaluation Engine is also referred as "Query execution engine". Query Evaluation Engine first interprets the SQL commands to access the data from the database and then it returns the answer to that query.

b) Write the SQL queries for following EMP table. Emp (empno, deptno, ename, salary, designation, city.)

i) Display average salary of all employees.

ii) Display names of employees who stay in Mumbai or Pune.

iii) Set the salary of employee 'Ramesh' to 50000.

iv) Display names of employees whose salaries are less than 50000.

v) Remove the Record of employees whose deptno is 10.

vi) Remove the column deptno from EMP table.

Ans. i. select avg(salary) from emp;

ii. select ename from emp where city='Mumbai' or city='Pune';

iii. update emp set salary=50000 where ename='Ramesh';

iv. select ename from emp where salary<50000;

v. delete from emp where deptno=10;

vi. alter table emp drop column deptno;

c) Write and Explain the syntax for creating, Altering and dropping the sequence.

Ans. Syntax for creating sequence:

CREATE SEQUENCE sequence_name

START WITH initial_value

INCREMENT BY increment_value

MINVALUE minimum value

MAXVALUE maximum value

CYCLE|NOCYCLE ;

sequence_name: Name of the sequence.

initial_value: starting value from where the sequence starts.

Initial_value should be greater than or equal to minimum value and less than equal to maximum value.

increment_value: Value by which sequence will increment itself.

Increment_value can be positive or negative.

minimum_value: Minimum value of the sequence.

maximum_value: Maximum value of the sequence.

cycle: When sequence reaches its set_limit it starts from beginning.

nocycle: An exception will be thrown if sequence exceeds its max_value.

Example:

CREATE SEQUENCE sequence_1

start with 1

increment by 1

minvalue 0

maxvalue 100

cycle;

Alter sequence:

Syntax:

alter sequence <sequence_name> maxvalue <number>;

Alter sequence can change the maxvalue in the sequence created.

Dropping sequence:

Syntax:

drop sequence <sequence_name>;

To drop the sequence the DROP command is used.

a) Write SQL queries for following. Consider table stud (roll no, name, sub1, sub2, sub3)

i) Display names of student who got minimum mark in sub1.

ii) Display names of students who got above 40 marks in sub2.

iii) Display count of Students failed in sub2.

iv) Display average marks of sub1 of all students.

v) Display names of students whose name start with 'A' by arranging them in ascending order of sub1 marks.

vi) Display student name whose name ends with h' and subject 2 marks are between 60 to 75.

Ans. i. select name from stud where sub1= (select min(sub1) from stud);

ii. select name from stud where sub2>40;

iii. select count(*) from stud where sub2<40;

iv. select avg(sub1) from stud;

v. select name from stud where name like 'A%' order by sub1;

vi. select name from stud where name like '%h' and sub2 between 60 and 75;

b) Write a PL/SQL code to check whether specified employee is present in Emp table or not. Accept empno from user. If employee does not exist display message using exception handling.

Ans. PL/SQL Program:

```
declare
no emp.empno%type;
begin
no:=&no;
select empno into no from emp where empno=no;
dbms_output.put_line('Empno is present: ' || no);
exception
when no_data_found then
dbms_output.put_line('Empno not present');
end;
/
```

c) Write SQL queries for following.

i) Create user named 'user1' having Password '1234

ii) Assign 'insert' and update' Privilege to 'user1'.

ii) Remove update Privilege assigned to the user1.

iv) Assign the resource Permission to user1.

Ans. i. create user user1 identified by 1234;

ii. grant insert, update on system.emp to user1;

iii. revoke update on system.emp from user1;

iv. grant create session, unlimited tablespace, create table to user1;